Math 10C — Unit 9: Assignments

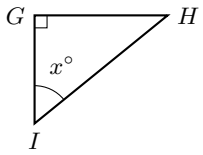
The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

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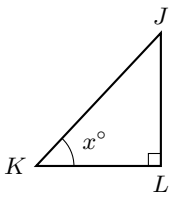
1. For each triangle below, state the opposite side, the adjacent side and the hypotenuse (as pairs of vertex letters) relative to the marked angle $x°$, then write $\sin x°$, $\cos x°$ and $\tan x°$ as ratios of those side names.

(a)



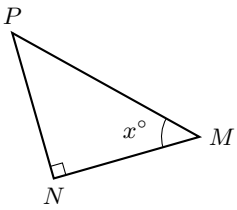
opp, adj, hyp =

(b)



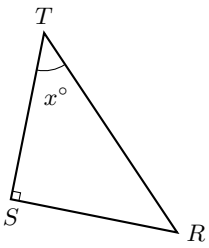
opp, adj, hyp =

(c)



opp, adj, hyp =

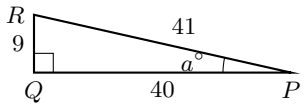
(d)



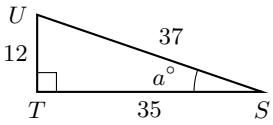
opp, adj, hyp =

2. Mark the hyp, opp and adj for the angle $a°$ on each triangle below, then state $\sin a°$, $\cos a°$ and $\tan a°$ as fractions in simplest form.

(a)

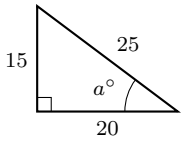


(b)

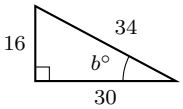


3. Determine, as a fraction in simplest form, the value of the trigonometric ratio indicated.

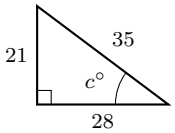
(a) $\cos a°$



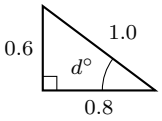
(b) $\sin b°$



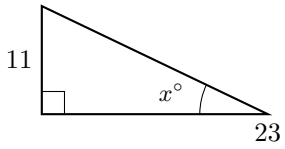
(c) $\tan c°$



(d) $\sin d°$



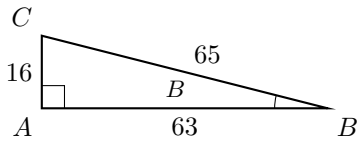
4. In the right triangle below, the two legs are 11 and 23 units.



(a) Use the Pythagorean theorem to find the hypotenuse, rounded to the nearest hundredth.

(b) Determine $\sin x°$, $\cos x°$ and $\tan x°$ as decimals rounded to the nearest hundredth.

5. In the right triangle below, determine the exact values of $\sin B$, $\cos B$ and $\tan B$ as fractions in simplest form.

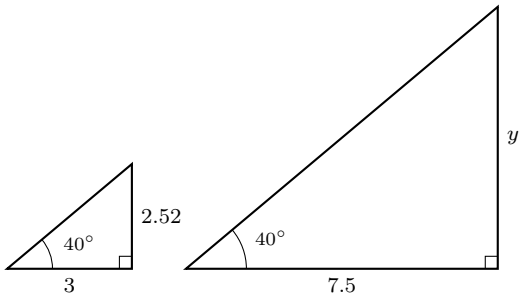


6. Sketch and label each right triangle from the ratio given, then determine the other two ratios as fractions in simplest form.

(a) In $\triangle ABC$, $\angle C = 90°$ and $\sin A = \dfrac{20}{29}$. Determine $\cos A$ and $\tan A$.

(b) In $\triangle DEF$, $\angle F = 90°$ and $\cos D = 0.6$. Determine $\sin D$ and $\tan D$.

7. Two right triangles each have a 40° angle. In the smaller triangle, the leg adjacent to the 40° angle is 3 and the leg opposite it is 2.52. In the larger triangle, the adjacent leg is 7.5 and the opposite leg is y.



(a) Explain why the two triangles are similar.

(b) Use the scale factor from the smaller triangle to the larger one to find y.

(c) Calculate $\dfrac{\text{opp}}{\text{adj}}$ in each triangle. Explain, using similarity, why $\tan 40°$ has the same value in both triangles, and compare with the calculator value of $\tan 40°$.

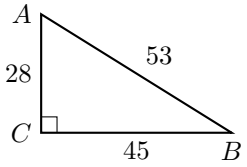
8. Explain why the size of a right triangle does not change the value of $\sin 31°$. Use the triangles of Example 1 in your explanation.

9. *(Multiple Choice)* In right triangle ABC, the right angle is at C, $AB = 37$ units, $AC = 35$ units and $BC = 12$ units. The values of $\cos B$ and $\sin B$, respectively, are

A. $\frac{12}{37}$ and $\frac{35}{37}$ B. $\frac{35}{37}$ and $\frac{12}{37}$

C. $\frac{12}{35}$ and $\frac{35}{12}$ D. $\frac{35}{12}$ and $\frac{12}{35}$

10. *(Multiple Choice)* For the right triangle ABC shown, only one of the following ratios is correct. The correct ratio is



A. $\sin A = \frac{28}{53}$ B. $\cos A = \frac{45}{53}$

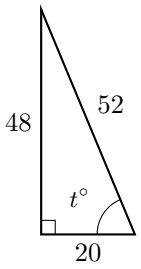
C. $\tan B = \frac{28}{45}$ D. $\sin B = \frac{45}{28}$

11. *(Multiple Choice)* In a right triangle, $\sin y° = \frac{9}{11}$. A student claims the side opposite $y°$ is 9 and the hypotenuse is 11. This claim

A. is always true B. is always false

C. may be true or false D. depends on the value of $y°$

12. *(Numerical Response)* In the diagram, to the nearest tenth, the value of $\frac{\sin t°}{\cos t°}$ is _____.



Record your answer left to right. | | | | |

Lesson 2 Assignment *Show your work.*

1. Use a calculator (in degree mode) to determine the value of each trigonometric ratio, to four decimal places.
(a) $\sin 72° =$ (b) $\tan 40° =$ (c) $\cos 15° =$ (d) $\tan 82° =$ (e) $\sin 42.3° =$ (f) $\cos 77.4° =$

2. Explain, using diagrams, why the sine ratio for an angle of 26° is smaller than the sine ratio for an angle of 52°.

3. Explain, using diagrams, why the cosine ratio for an angle of 25° is greater than the cosine ratio for an angle of 65°.

4. Explain, using diagrams, why the tangent ratio for an angle of 12° is smaller than the tangent ratio for an angle of 55°.

5. Without using a calculator, state which ratio in each part has the greater value.
(a) $\sin 63°,\ \sin 29°$ (b) $\cos 71°,\ \cos 38°$ (c) $\tan 48°,\ \tan 83°$

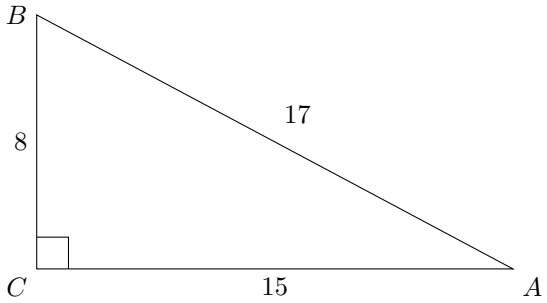
(d) $\sin 34°,\ \cos 34°$ (e) $\sin 80°,\ \cos 80°$ (f) $\sin 45°,\ \cos 45°,\ \tan 45°$

6. (a) Complete the table, rounding to two decimal places.

Angle $x°$	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°
$\cos x°$										
$\tan x°$										

(b) Complete the statements: as the angle $x°$ increases from 0° to 90°, $\cos x°$ _____ from _____ to _____, and $\tan x°$ _____ from _____, passing 1 at _____.

7. Triangle ABC shown below has a right angle at C. (a) Referring to triangle ABC, write the rational number that represents each indicated trigonometric ratio.



$\sin A =$ $\cos A =$ $\tan A =$

$\sin B =$ $\cos B =$ $\tan B =$

(b) Which pairs of trigonometric ratios above have the same value?

(c) How do the values of $\tan A$ and $\tan B$ relate to each other?

8. *(Multiple Choice)* For *every* acute angle $x°$, which statement is true?
A. $\sin x°$ can be greater than 1
B. $\cos x°$ is less than 1
C. $\tan x°$ is less than 1
D. $\sin x°$ is less than $\cos x°$

9. *(Numerical Response)* To the nearest tenth, the value of $\dfrac{\cos 60° + \sin 60°}{\tan 60°}$ is _____.

Record your answer left to right. ☐☐☐☐

Extension — Tangent Near $90°$ and the Cofunction Relationship

10. *(Extension)* Use your table in Question 6 and Example 4(d). Complete the statement: as the angle $x°$ increases toward $90°$, $\tan x°$ _____. Explain why, using $\tan x° = \dfrac{\text{opp}}{\text{adj}}$.

11. *(Extension)* (a) Draw a right triangle PQR in which $\angle P = 90°$ and $\angle Q = x°$.

(b) Write an expression for the measure of $\angle R$ in terms of x.

(c) Explain, using ratios, why $\cos x° = \sin(90 - x)°$.

(d) Complete the following, using an angle between $0°$ and $90°$.
(i) $\cos 68° = \sin$ _____

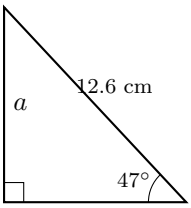
(ii) $\cos 34° = \sin$ _____

12. *(Multiple Choice; Extension)* Triangle STU is right-angled at S, with $\tan T = k$. Three statements are made about the trigonometric ratios in triangle STU.
Statement 1. $\sin T = \cos U$
Statement 2. $\tan U = k$
Statement 3. $\cos T = \sin U$
How many of these statements are true?
A. 0 B. 1 C. 2 D. 3

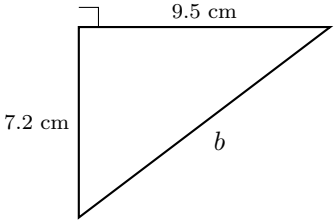
Lesson 3 Assignment *Show your work.*

1. Calculate, to the nearest tenth, the length of the indicated side in each triangle.

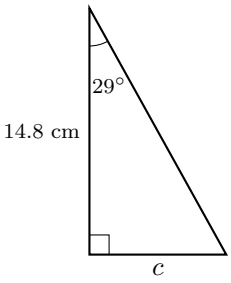
(a)



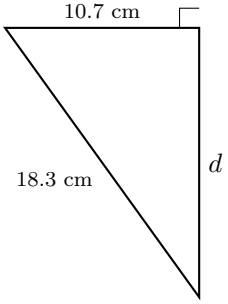
(b)



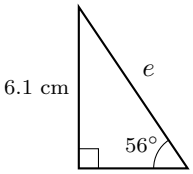
(c)



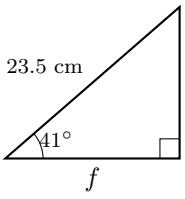
(d)



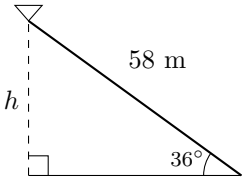
(e)



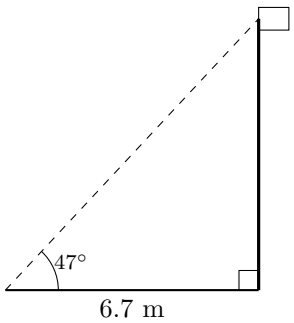
(f)



2. The string of a kite is 58 m long and makes an angle of 36° with the ground. Calculate, to the nearest metre, the vertical height, h, of the kite above the ground.

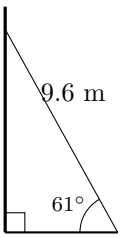


3. Use the measurements in the diagram to determine the height of the flagpole to the nearest tenth of a metre.



4. A guy wire 9.6 m long is attached to a pole and anchored to the ground, making an angle of 61° with the ground.

(a) How high up the pole, to the nearest tenth of a metre, does the wire attach?



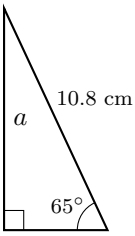
(b) Use trigonometry to determine, to the nearest tenth of a metre, the distance between the base of the pole and the point where the wire is anchored to the ground.

(c) Use your answer to (a) and the Pythagorean theorem to determine that same distance, to the nearest tenth of a metre.

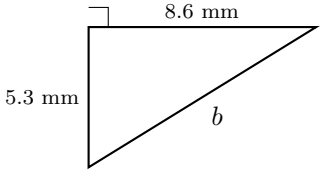
(d) Explain why the answers to (b) and (c) are not exactly the same.

5. Calculate, to the nearest tenth, the length of the indicated side in each triangle.

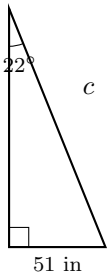
(a)



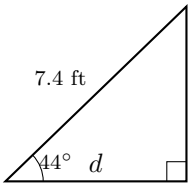
(b)



(c)

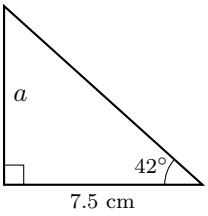


(d)

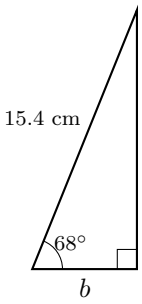


6. Calculate, to the nearest tenth of a centimetre, the length of the indicated side in each triangle.

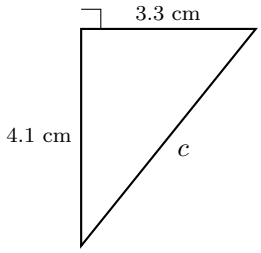
(a)



(b)

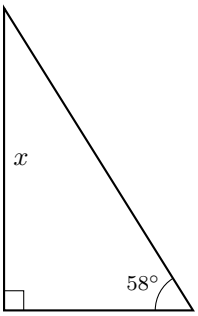


(c)

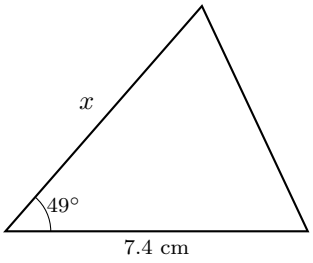


7. Explain why trigonometric ratios could not be used to calculate the side marked x in each of the following triangles.

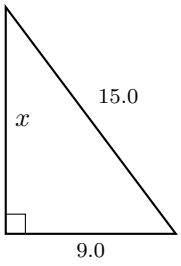
(a)



(b)



(c)

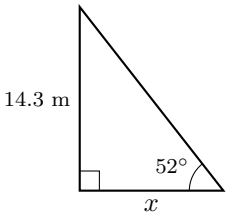


8. Determine, to the nearest tenth, the length of the hypotenuse of each triangle. Start by sketching and labelling the triangle.

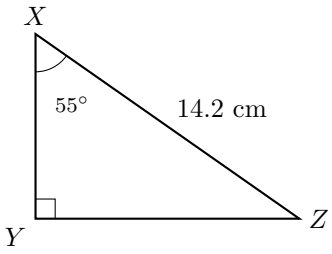
(a) Triangle STU in which angle $SUT = 90°$, angle $STU = 52°$, and $SU = 27.8$ mm.

(b) Triangle GHI in which angle $GHI = 90°$, angle $GIH = 41°$, and $HI = 9.3$ cm.

9. Determine the length of the leg x, to the nearest tenth of a metre.



10. From a point P on level ground, the line of sight to the top of a 24 m flagpole makes an angle of 38° with the ground. Sketch and label the right triangle, then determine how far P is from the base of the pole, to the nearest tenth of a metre.



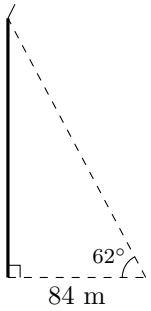
11. Calculate, to the nearest tenth of a centimetre, the lengths of XY and YZ in the triangle shown.

12. *(Multiple Choice)* Triangle DEF is right-angled at F. Angle $DEF = 41°$ and $DF = 13$ cm. The length of DE, in cm, is given by

A. $13 \sin 41°$ B. $\dfrac{13}{\cos 41°}$

C. $13 \tan 41°$ D. $\dfrac{13}{\cos 49°}$

13. *(Numerical Response)* A radio tower casts a shadow 84 m long. The angle between the ground and the line from the tip of the shadow to the top of the tower is 62°. To the nearest metre, the height of the tower is _____.



Record your answer left to right. | | | | |

14. *(Numerical Response)* In right triangle ABC, angle $ABC = 90°$, angle $BAC = 58°$, and $AC = 34$ units. To the nearest whole number, the perimeter of the triangle is _____.

Record your answer left to right. | | | | |

Lesson 4 Assignment *Show your work.*

1. In each case, use a calculator to determine the measure of the acute angle to the nearest degree.

(a) $\sin a^\circ = 0.8192$, $a^\circ =$ (b) $\cos b^\circ = 0.3907$, $b^\circ =$ (c) $\tan P = 0.9004$, $\angle P =$ (d) $\sin C = \frac{2}{5}$, $\angle C =$

(e) $\sin s^\circ = \frac{\sqrt{3}}{2}$, $s^\circ =$ (f) $\tan y^\circ = 1$, $y^\circ =$ (g) $\cos A = \frac{1}{2}$, $\angle A =$ (h) $\sin X = \frac{\sqrt{2}}{2}$, $\angle X =$

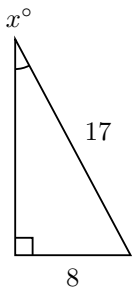
2. In each case, use a calculator to determine the indicated acute angle to the nearest tenth of a degree.

(a) $\sin A = 0.5934$ (b) $\cos X = 0.2076$ (c) $\tan P = 0.82$

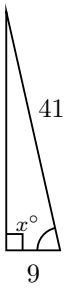
(d) $\cos e^\circ = \frac{9}{41}$ (e) $\tan M = 25$ (f) $\sin R = \frac{7}{9}$

3. In each case, calculate the measure of the indicated angle to the nearest degree.

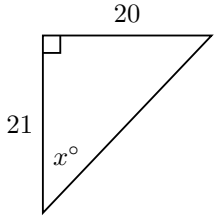
(a)



(b)

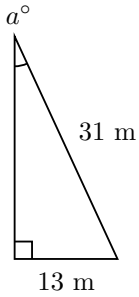


(c)

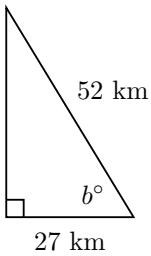


4. In each case, calculate the measure of the indicated angle to the nearest tenth of a degree.

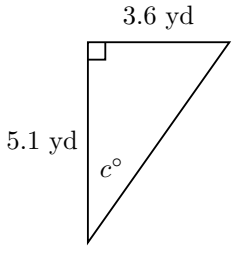
(a)



(b)

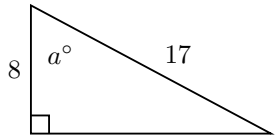


(c)

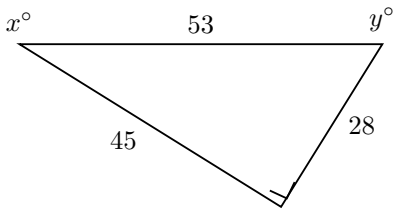


5. In each case, calculate the measure of the indicated angle(s) to the nearest degree.

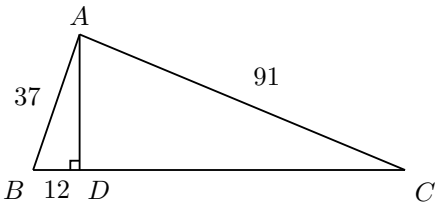
(a)



(b)



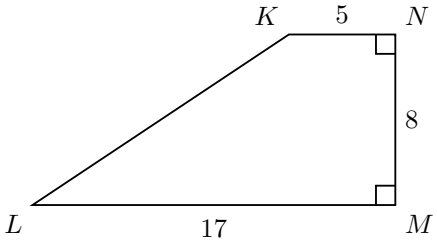
6. The diagram shows two right triangles with a common side AD.



(a) Use the Pythagorean Theorem to calculate the length of AD.

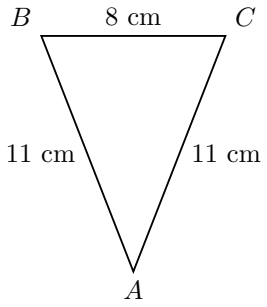
(b) Determine, to the nearest degree, the measure of angle BCA.

7. Determine the measure of angle LKN to the nearest degree.

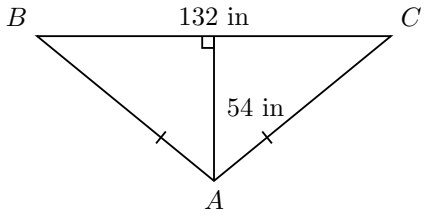


8. A pylon 31.5 m high casts a shadow 52.6 m long. Determine the angle of elevation of the sun to the nearest tenth of a degree.

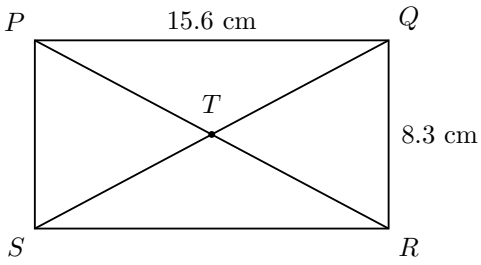
9. In each case, calculate the size of angle BAC to the nearest degree.
(a)



(b)

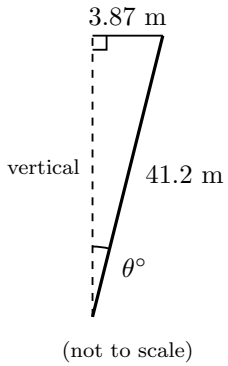


10. In rectangle $PQRS$, $PQ = 15.6$ cm and $QR = 8.3$ cm, and the diagonals meet at T. Determine the measure of angle PTQ to the nearest degree.



11. The term "% grade" describes the slope of a road: a road with an 11% grade rises 11 m vertically for every 100 m of horizontal distance. Calculate, to the nearest degree, the angle a road with an 11% grade makes with the horizontal.

12. A leaning clock tower has drifted from the vertical. Using the measurements in the sketch, determine, to the nearest hundredth of a degree, the tower's angle of lean from the vertical.



(not to scale)

13. *(Multiple Choice)* A wheelchair ramp has a vertical rise of 9 cm for every 76 cm of horizontal run. To the nearest degree, the angle between the ramp and the floor is

A. 6° B. 7° C. 83° D. 84°

14. *(Numerical Response)* A tourist atop a bridge tower spots a boat in the water below. The angle of depression of the boat from the tourist is 52°. At the same moment, the angle of elevation of the tourist from the boat, to the nearest degree, is _____.

Record your answer left to right. | | | | |

15. *(Numerical Response)* A window washer on a platform 48 m above the street looks down at a parked bus that is 65 m from the base of the building. To the nearest tenth of a degree, the angle of depression of the bus from the window washer is _____.

Record your answer left to right. | | | | |

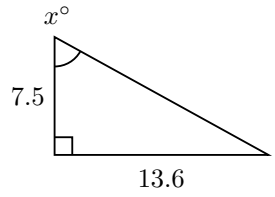
16. *(Numerical Response)* A cargo drone descends at a constant angle of depression and travels 340 m along its flight path while descending. When it levels off, it has dropped a vertical distance of 210 m. To the nearest tenth of a degree, the angle of depression of the descent is _____.

Record your answer left to right. | | | | |

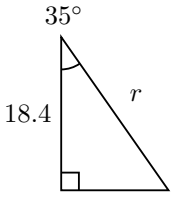
Lesson 5 Assignment *Show your work.*

1. In each case, calculate the indicated measure. Give angles to the nearest degree and sides to the nearest tenth.

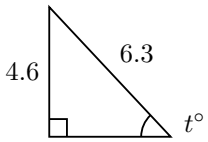
(a)



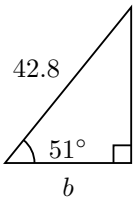
(b)



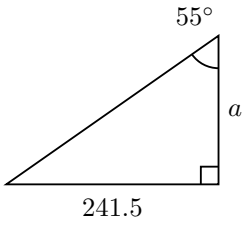
(c)



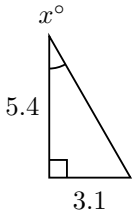
(d)



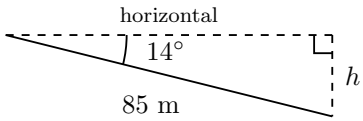
(e)



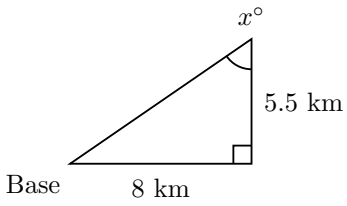
(f)



2. A drainage pipe slopes downward at an angle of 14° to the horizontal. If the pipe is 85 m long, calculate, to the nearest metre, how much lower its bottom end is than its top end.



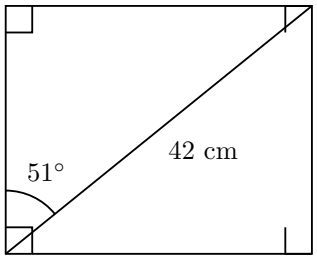
3. A drone flies 8 km due east from base camp to a checkpoint, then 5.5 km due north to a second checkpoint, then flies directly back to base camp.



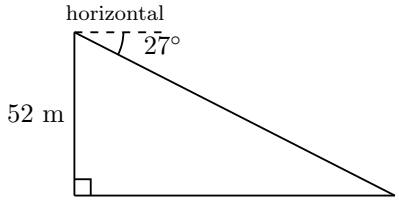
(a) Calculate the measure of angle $x°$ to the nearest degree.

(b) Calculate, to the nearest tenth of a km, the total distance the drone travels.

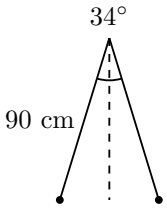
4. The diagonal of a rectangle is 42 cm long and makes an angle of 51° with the shorter side. Determine the length of the shorter side and the longer side, each to the nearest tenth of a cm.



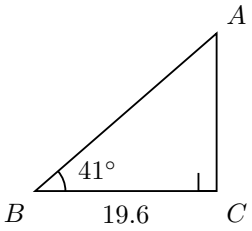
5. From the top of an observation tower, the angle of depression to a parked car is 27°. The tower is 52 m tall. Calculate, to the nearest metre, the distance from the base of the tower to the car.



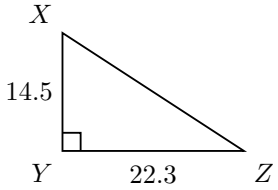
6. A pendulum 90 cm long swings through an angle of 34°. Calculate, to the nearest tenth of a cm, the distance between the two extreme positions of the pendulum bob.



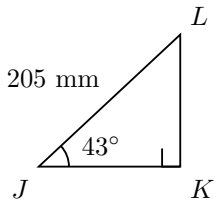
7. Solve triangle ABC, giving angles to the nearest degree and sides to the nearest tenth.



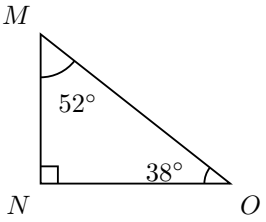
8. Solve triangle XYZ, giving angles to the nearest degree and sides to the nearest tenth.



9. Solve triangle JKL, in which angle $JKL = 90°$, angle $KJL = 43°$ and $JL = 205$ mm. Give all answers to the nearest whole number.



10. Why is it not possible to solve triangle MNO, in which angle $MNO = 90°$, angle $NMO = 52°$ and angle $MON = 38°$?



11. *(Multiple Choice)* A straight zipline cable is 120 m long and slopes down at an angle of 9° below the horizontal. To the nearest tenth of a metre, the vertical drop from the top of the cable to the bottom is

A. 18.8 m B. 19.0 m
C. 118.5 m D. 121.5 m

12. *(Numerical Response)* An airplane flying at an altitude of 1.8 km begins a straight descent toward the start of a runway. The angle of depression from the plane to the start of the runway is 3°. To the nearest tenth of a kilometre, the horizontal distance from the plane to the start of the runway is _____ km.

Record your answer left to right. | | | | |

13. *(Numerical Response)* A children's slide is 3.4 m long and reaches a height of 2.1 m. To the nearest tenth of a degree, the angle the slide makes with the ground is _____°.

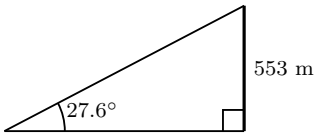
Record your answer left to right. | | | | |

14. *(Numerical Response)* In right triangle PQR, $\angle Q = 90°$, $\angle P = 33°$ and the hypotenuse $PR = 48.0$ cm. To the nearest square centimetre, the area of the triangle is _____ cm^2.

Record your answer left to right. | | | | |

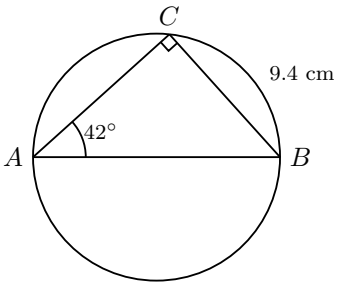
Lesson 6 Assignment *Show your work.*

1. The CN Tower is 553 m tall. On a sunny afternoon it casts a shadow, and a tourist standing at the tip of the shadow measures the angle of elevation of the sun to be 27.6°. Determine, to the nearest metre, the length of the shadow.



2. In the diagram, AB is a diameter of a circle. Angle $CAB = 42°$ and $CB = 9.4$ cm.

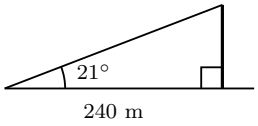
(a) Explain why angle $ACB = 90°$.



(b) Determine the radius of the circle to the nearest tenth of a centimetre.

3. A boat is 240 m from the base of a lighthouse cliff. The angle of elevation from the boat to the top of the cliff is 21°.

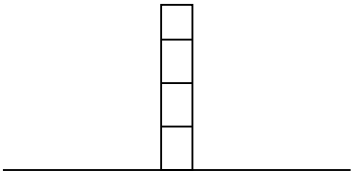
(a) Determine the height of the cliff to the nearest metre.



(b) If the boat then sails 60 m closer to the cliff, determine, to the nearest degree, the new angle of elevation of the cliff top from the boat.

4. A sculpture pole is being erected in a 1.5 m deep hole. Two guy wires attached to the top of the pole are used to pull it upright, anchored into the ground on *opposite* sides of the pole until the hole is filled. Each wire is 22 m long and is anchored into the ground 14 m horizontally from the centre of the pole.

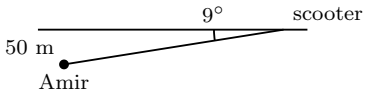
(a) Complete the sketch to illustrate this situation.



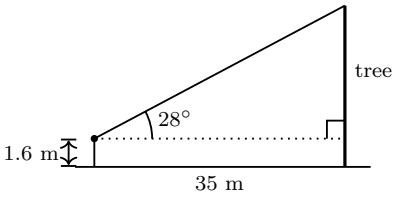
(b) Calculate, to the nearest tenth of a degree, the angle of elevation of the top of the pole from the point where a wire is anchored into the ground.

(c) Determine, to the nearest tenth of a metre, the length of the pole before it was placed in the ground.

5. Amir is standing 50 m south of a straight east-west highway. He looks to his right and notices a motor-scooter approaching along the highway; at that instant, the angle between the highway and his line of sight to the scooter is 9°. Fifteen seconds later, the scooter is directly north of Amir. Calculate the scooter's speed, to the nearest km/h.



6. Leah's eyes are 1.6 m above the ground. She stands 35 m from the base of a tree and measures the angle of elevation of the top of the tree, from her eye level, as 28°. Determine the height of the tree to the nearest tenth of a metre.



7. A lifeguard's eyes are 6.2 m above the water, directly above the base of her tower. She spots a swimmer at an angle of depression of 9°. The swimmer swims straight toward the base of the tower at 0.8 m/s.

(a) How far is the swimmer from the base of the tower, to the nearest tenth of a metre?

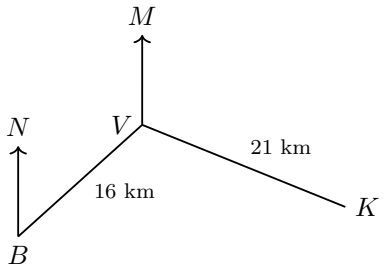
(b) How long, to the nearest second, until the swimmer reaches the base of the tower?

8. *(Numerical Response)* Using a clinometer held at eye level, 1.5 m above the ground, a student standing 30 m from the base of a flagpole reads an angle of elevation of 22° to the top of the pole. To the nearest tenth of a metre, the height of the flagpole is _____ m.

Record your answer left to right. | | | | |

Extension — Bearings

9. *(Extension)* A rescue helicopter flies from a base station, B, on a bearing of 048° for 16 km to a checkpoint, V. At V it changes course and flies on a bearing of 112° for 21 km to a platform, K. BN and VM represent north lines at B and V.



(a) On the diagram, mark the sizes of angle NBV and angle MVK.

(b) Determine, to the nearest 0.1 km, how far K is east of B.

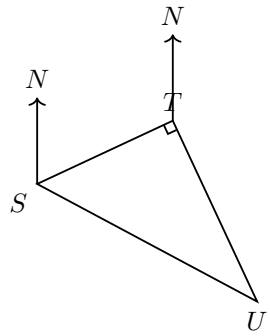
(c) Determine, to the nearest 0.1 km, how far K is north of B.

(d) Determine, to the nearest 0.1 km, the direct distance from B to K.

(e) Determine the three-figure bearing of K from B.

Use the following information to answer Questions 10 and 11 (Extension).

A drone starts from a station S and flies 150 m on a bearing of 065° to point T. At T, the drone makes a 90° turn and flies 200 m on a bearing of 155° to point U. At U it changes course and flies directly back to S.



10. *(Multiple Choice)* The direct distance, in m, from U to S is
A. 200 B. 250 C. 300 D. 350

11. *(Multiple Choice)* The three-figure bearing the drone uses to fly from U back to S is
A. 062° B. 118° C. 242° D. 298°

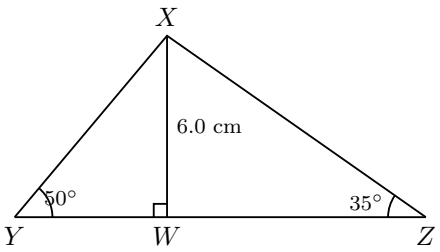
Lesson 7 Assignment *Show your work.*

1. *(Warm-up — fill in the frame.)* In the diagram, XW is the altitude from X to YZ, $XW = 6.0$ cm, $\angle Y = 50°$ and $\angle Z = 35°$. Determine YZ to the nearest tenth of a centimetre.
The shared side is _____.

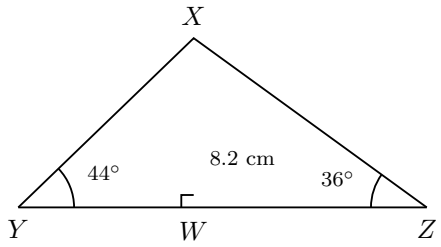
In right triangle XWY: $\tan 50° = \frac{6.0}{\text{_____}}$, so $YW =$ _____.

In right triangle XWZ: $\tan 35° = \frac{6.0}{\text{_____}}$, so $WZ =$ _____.

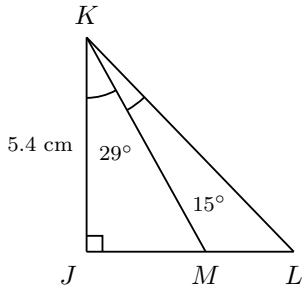
$YZ = YW + WZ =$ _____.



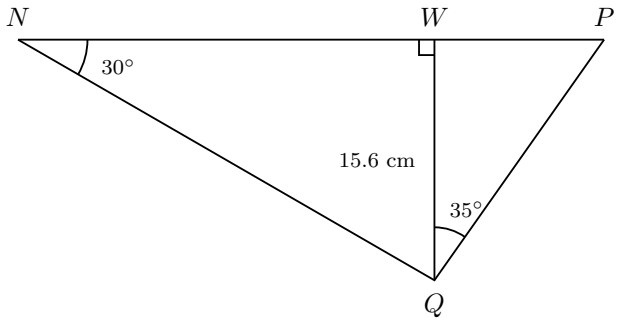
2. Determine the length of the requested segment, to the nearest 0.1 cm, in each diagram.
(a) XW is the altitude from X to YZ. Determine YZ.



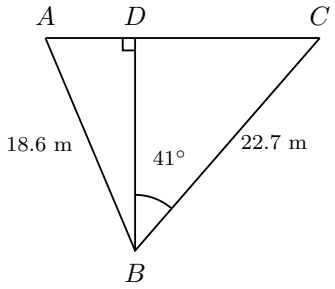
(b) $\angle JKM = 29°$ and $\angle MKL = 15°$, where $JK \perp JL$. Determine ML.



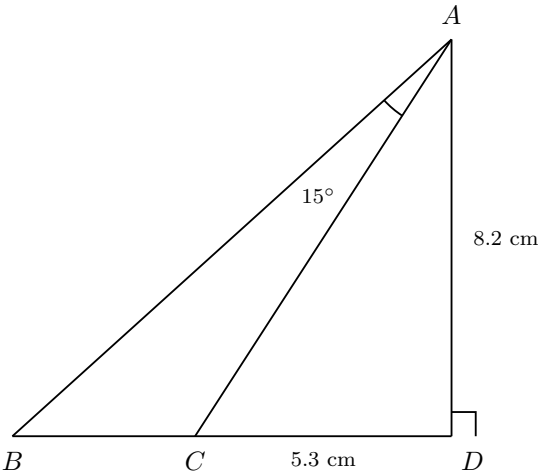
(c) QW is perpendicular to NP and $\angle WQP = 35°$. Determine NP.



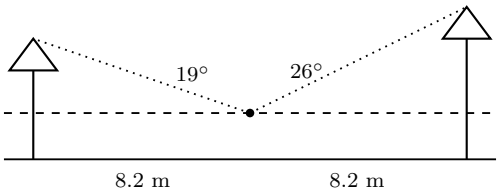
3. Determine the measure of $\angle ABC$, to the nearest degree, in each diagram.
(a) BD is the altitude from B to AC, and $\angle DBC = 41°$.



(b) $AD \perp BD$, C lies on BD, and $\angle CAB = 15°$.



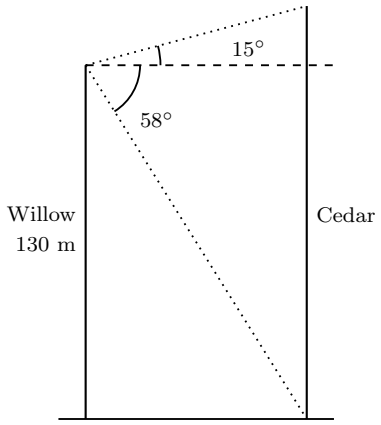
4. Two trees in a park are 16.4 m apart. An observer, whose eyeline is 1.75 m above the ground, stands halfway between the trees. The angles of elevation of the tops of the trees from the observer's eyeline are 19° and 26°.



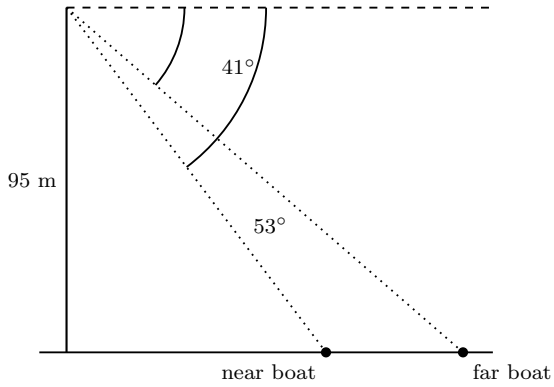
(a) Why must 1.75 m be added to each computed rise?

(b) Determine the height of each tree, to the nearest 0.01 m.

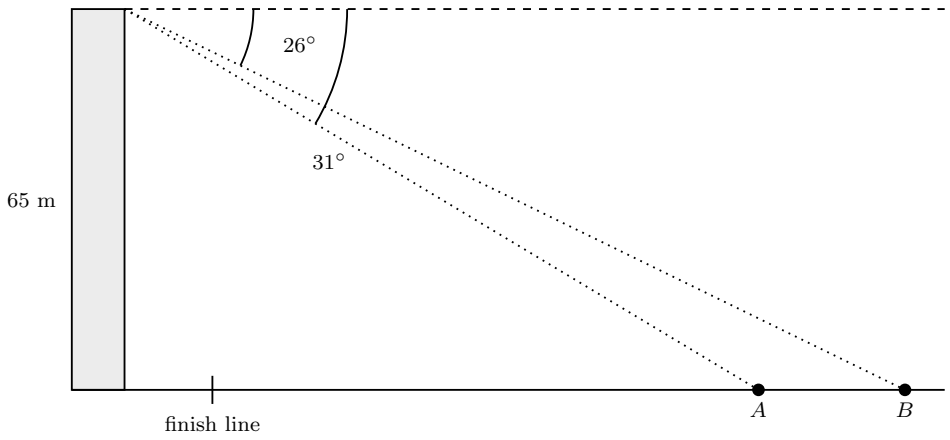
5. Willow Tower is 130 m tall. From the top of Willow Tower, the angle of depression to the foot of Cedar Tower is 58°, and the angle of elevation to the top of Cedar Tower is 15°. Determine the height of Cedar Tower to the nearest tenth of a metre.



6. From the top of a cliff 95 m high, an observer sees two boats heading toward shore, one directly behind the other. The angle of depression to the farther boat is 41°, and the angle of depression to the nearer boat is 53°. Determine the distance between the boats, to the nearest metre.



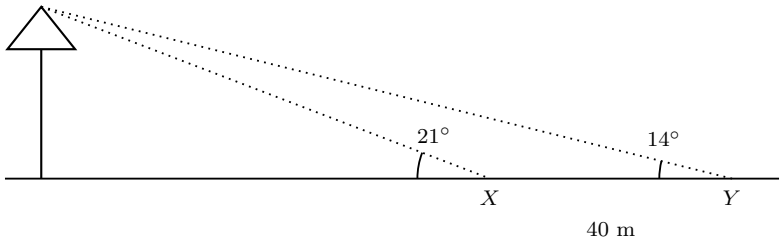
7. An apartment window is 65 m above the ground and 15 m behind the finish line of a race. Two runners, A and B, are heading toward the finish line. From the window, the angle of depression to runner A is 31°, and the angle of depression to runner B is 26°.



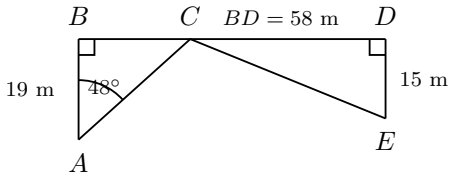
(a) Determine the distance between the runners, to the nearest metre.

(b) A runs at a constant 4.2 m/s and B runs at a constant 4.8 m/s. Which runner reaches the finish line first?

8. Standing at point X directly across a river from a tall tree, Jenny measures the angle of elevation to the top of the tree as 21°. She then walks 40 m straight back from the river to point Y and measures the angle of elevation to the same treetop as 14°. Determine the height of the tree to the nearest metre. *(Use the method of Example 4.)*



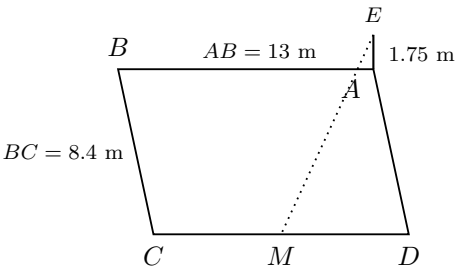
9. *(Numerical Response)* In the diagram, $AB = 19$ m, $DE = 15$ m, $BD = 58$ m, $AB \perp BD$, $ED \perp BD$, and $\angle BAC = 48°$. To the nearest degree, the measure of $\angle CED$ is _____.

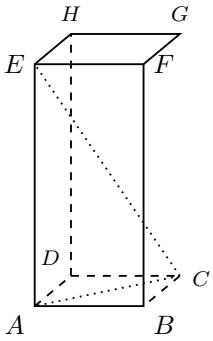


Record your answer left to right. | | | | |

Extension — Three Dimensions

10. *(Extension)* In the diagram, $ABCD$ represents a rectangular playing field, drawn here at a slight angle for perspective. A referee stands at corner A, and at a certain time of day the tip of the referee's shadow lands exactly at M, the midpoint of CD. The referee is 1.75 m tall, $AB = 13$ m, and $BC = 8.4$ m. Determine the measure of $\angle EMA$ (the angle of elevation of the sun), to the nearest degree.



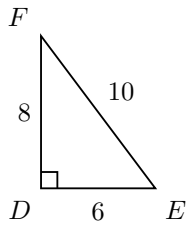


11. *(Multiple Choice; Extension)* A rectangular prism has base dimensions $AB = 9$ m and $BC = 12$ m, and height $AE = 20$ m, as shown (E is directly above A, and similarly for the other top vertices). The tangent of $\angle ACE$ (the angle between diagonal CE and the base plane $ABCD$) is

A. $\frac{3}{4}$ B. $\frac{4}{3}$ C. $\frac{5}{4}$ D. 1

Lesson 8 Assignment *Show your work.*

1. Triangle DEF has a right angle at D, with $DE = 6$, $DF = 8$, and $EF = 10$.

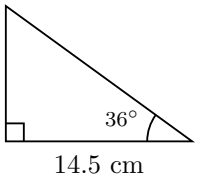


(a) Relative to angle E, name the side opposite E and the side adjacent to E.

(b) Write $\sin E$, $\cos E$ and $\tan E$ as fractions in lowest terms.

(c) Without further calculation, state the value of $\cos F$.

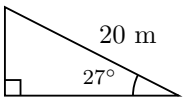
2. In the right triangle below, one acute angle measures 36° and the side adjacent to it measures 14.5 cm.



(a) Determine the length of the side opposite the 36° angle, to the nearest tenth of a centimetre.

(b) Determine the length of the hypotenuse, to the nearest tenth of a centimetre.

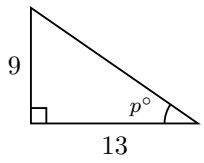
3. In the right triangle below, one acute angle measures 27° and the hypotenuse measures 20 m.



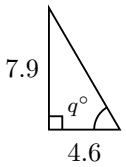
(a) Determine the length of the side opposite the 27° angle, to the nearest tenth of a metre.

(b) Determine the length of the side adjacent to the 27° angle, to the nearest tenth of a metre.

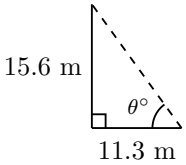
4. The leg opposite angle p° measures 9 and the leg adjacent to p° measures 13. Determine the measure of angle p°, to the nearest degree.



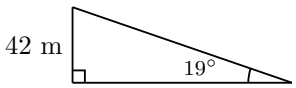
5. The leg adjacent to angle q° measures 4.6 and the leg opposite q° measures 7.9. Determine the measure of angle q°, to the nearest tenth of a degree.



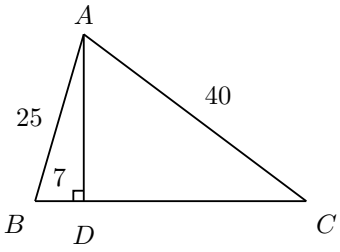
6. A tree casts a shadow 11.3 m long while the tree itself is 15.6 m tall. Determine, to the nearest tenth of a degree, the angle of elevation of the sun.



7. From the top of a cliff 42 m high, the angle of depression of a boat on the water is 19°. Determine, to the nearest tenth of a metre, the horizontal distance from the base of the cliff to the boat.



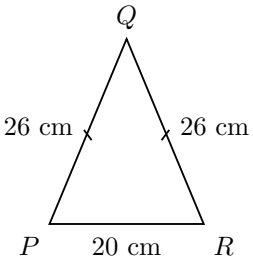
8. Two right triangles share the common side AD, with D on BC between B and C. $AB = 25$, $BD = 7$, and $AC = 40$.



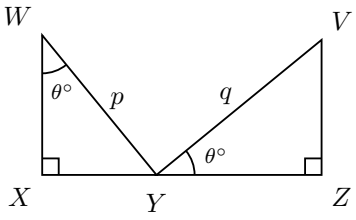
(a) Use the Pythagorean Theorem to calculate the length of AD.

(b) Determine, to the nearest degree, the measure of angle BAC.

9. Isosceles triangle PQR has $PQ = RQ = 26$ cm and base $PR = 20$ cm. Determine, to the nearest degree, the measure of $\angle PQR$.

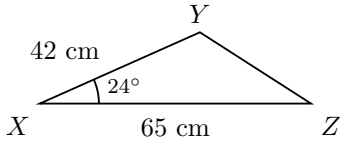


10. *(Multiple Choice)* In the diagram below, right angles occur at X and Z, $WY = p$, $YV = q$, $\angle YWX = \theta°$, and $\angle VYZ = \theta°$. The length of XZ is



A. $p \sin\theta° + q \cos\theta°$
B. $p \cos\theta° + q \sin\theta°$
C. $(p + q) \cos\theta°$
D. $(p + q) \sin\theta°$

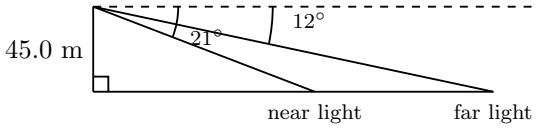
11. *(Numerical Response)* In triangle XYZ, $\angle X = 24°$, $XY = 42$ cm, and $XZ = 65$ cm.



The area of triangle XYZ, to the nearest square centimetre, is _____ cm^2.

Record your answer left to right. | | | | |

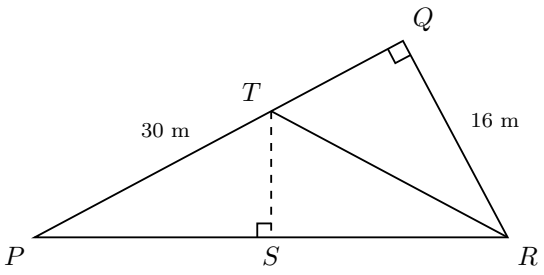
12. *(Numerical Response)* From a control tower window 45.0 m above the ground, the angles of depression of two runway lights, which are in a direct line from the tower, are 12° and 21° respectively.



To the nearest tenth of a metre, the distance between the two runway lights is _____ m.

Record your answer left to right. | | | | |

13. *(Written Response — 5 marks)* In the diagram, $\angle PQR = 90°$, $PQ = 30$ m, $QR = 16$ m, and T is a point on PQ such that $PT = TR$. S is the foot of the perpendicular from T to PR.



• Sketch triangle PQR only and determine the exact length of PR. *(1 mark)*

• Explain why triangle QRT alone (without using the condition $PT = TR$) does not give enough information to determine the measure of $\angle QRT$. *(1 mark)*

• Calculate the measure of $\angle QPR$, and use it to determine the measure of $\angle QRT$, to the nearest degree. *(1 mark)*

• Explain why S is the midpoint of PR, then use $\triangle PST$ to calculate the length of PT, to the nearest tenth of a metre. *(2 marks)*

14. Evaluate (a)–(c) to four decimal places, and determine the acute angle in (d)–(f) to the nearest degree.

(a) $\sin 37° =$ _____ (b) $\cos 71° =$ _____ (c) $\tan 54° =$ _____

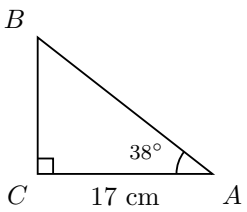
(d) $\sin\theta° = 0.5150$
$\theta° =$ _____

(e) $\cos\theta° = 0.2079$
$\theta° =$ _____

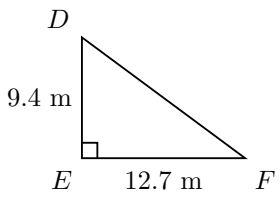
(f) $\tan\theta° = 3.2709$
$\theta° =$ _____

15. Solve each right triangle completely: state every side length to the nearest tenth and every angle to the nearest tenth of a degree.

(a) Triangle ABC has $\angle C = 90°$, $\angle A = 38°$ and $AC = 17$ cm.



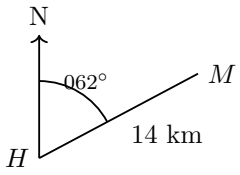
(b) Triangle DEF has $\angle E = 90°$, $DE = 9.4$ m and $EF = 12.7$ m.



16. A surveyor whose eye level is 1.7 m above the ground stands 48 m from the base of a grain elevator and measures the angle of elevation of its top as 31°. Determine the height of the grain elevator, to the nearest tenth of a metre.

Extension — Bearings and Three Dimensions

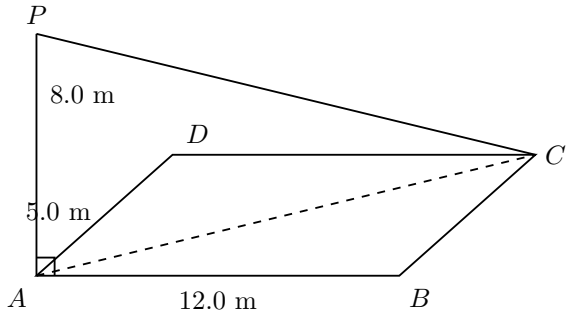
17. *(Extension)* A boat leaves harbour H and sails 14 km on a bearing of 062° to a marker M.



(a) Determine, to the nearest tenth of a kilometre, how far north of H and how far east of H the marker M is.

(b) From M the boat sails due south for 9.4 km to a point N. Determine the distance HN to the nearest tenth of a kilometre, and the three-figure bearing of N from H to the nearest degree.

18. *(Extension)* A vertical flagpole AP stands at corner A of a horizontal rectangular courtyard $ABCD$, where $AB = 12.0$ m, $AD = 5.0$ m and $AP = 8.0$ m.



(a) Determine the length of the diagonal AC, to the nearest tenth of a metre.

(b) Determine the angle of elevation of P from C, to the nearest tenth of a degree.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) opp= GH, adj= GI, hyp= HI; $\sin x^\circ = \frac{GH}{HI}$, $\cos x^\circ = \frac{GI}{HI}$, $\tan x^\circ = \frac{GH}{GI}$ (b) opp= LJ, adj= LK, hyp= KJ; $\sin x^\circ = \frac{LJ}{KJ}$, $\cos x^\circ = \frac{LK}{KJ}$, $\tan x^\circ = \frac{LJ}{LK}$ (c) opp= NP, adj= NM, hyp= MP; $\sin x^\circ = \frac{NP}{MP}$, $\cos x^\circ = \frac{NM}{MP}$, $\tan x^\circ = \frac{NP}{NM}$ (d) opp= SR, adj= ST, hyp= RT; $\sin x^\circ = \frac{SR}{RT}$, $\cos x^\circ = \frac{ST}{RT}$, $\tan x^\circ = \frac{SR}{ST}$. **2.** (a) $\sin a^\circ = \frac{9}{41}$, $\cos a^\circ = \frac{40}{41}$, $\tan a^\circ = \frac{9}{40}$ (b) $\sin a^\circ = \frac{12}{37}$, $\cos a^\circ = \frac{35}{37}$, $\tan a^\circ = \frac{12}{35}$. **3.** (a) $\frac{4}{5}$ (b) $\frac{8}{17}$ (c) $\frac{3}{4}$ (d) $\frac{3}{5}$. **4.** (a) hypotenuse ≈ 25.50 (b) $\sin x^\circ \approx 0.43$, $\cos x^\circ \approx 0.90$, $\tan x^\circ \approx 0.48$. **5.** $\sin B = \frac{16}{65}$, $\cos B = \frac{63}{65}$, $\tan B = \frac{16}{63}$. **6.** (a) $\cos A = \frac{21}{29}$, $\tan A = \frac{20}{21}$ (b) $\sin D = \frac{4}{5}$, $\tan D = \frac{4}{3}$. **7.** (a) Both have angles 90°, 40°, 50°, so they are similar. (b) $y = 6.3$ (c) 0.84 in both: the scale factor multiplies both legs, so their ratio is unchanged ($\tan 40^\circ \approx 0.839$). **8.** All right triangles with a 31° angle are similar; every side is multiplied by the same factor, so opp ÷ hyp does not change. **9.** A. **10.** C. **11.** C. **12.** 2.4.

Lesson 2 Assignment

1. (a) 0.9511 (b) 0.8391 (c) 0.9659 (d) 7.1154 (e) 0.6730 (f) 0.2181. **2.** With the hypotenuse the same in both triangles, the opposite side is shorter for 26° than for 52°; since $\sin = \frac{\text{opp}}{\text{hyp}}$, the sine ratio for 26° is smaller ($0.4384 < 0.7880$). **3.** With the hypotenuse the same in both triangles, the adjacent side is longer for 25° than for 65°; since $\cos = \frac{\text{adj}}{\text{hyp}}$, the cosine ratio for 25° is greater ($0.9063 > 0.4226$). **4.** From 12° to 55° the opposite side grows and the adjacent side shrinks, so $\tan = \frac{\text{opp}}{\text{adj}}$ is greater at 55° ($0.2126 < 1.4281$). **5.** (a) $\sin 63^\circ$ (b) $\cos 38^\circ$ (c) $\tan 83^\circ$ (d) $\cos 34^\circ$ (e) $\sin 80^\circ$ (f) $\tan 45^\circ$. **6.** (a) cos: 1.00, 0.98, 0.94, 0.87, 0.77, 0.64, 0.50, 0.34, 0.17, 0.00; tan: 0.00, 0.18, 0.36, 0.58, 0.84, 1.19, 1.73, 2.75, 5.67, undefined (b) $\cos x^\circ$ decreases from 1 to 0; $\tan x^\circ$ increases from 0, passing 1 at 45°. **7.** (a) $\sin A = \frac{8}{17}$, $\cos A = \frac{15}{17}$, $\tan A = \frac{8}{15}$, $\sin B = \frac{15}{17}$, $\cos B = \frac{8}{17}$, $\tan B = \frac{15}{8}$ (b) $\sin A = \cos B$ and $\cos A = \sin B$ (c) $\tan A$ and $\tan B$ are reciprocals. **8.** B. **9.** 0.8. **10.** (Extension) increases without bound (undefined at 90°). **11.** (Extension) (b) $(90 - x)^\circ$ (d)(i) $\sin 22^\circ$ (ii) $\sin 56^\circ$. **12.** (Extension) C.

Lesson 3 Assignment

1. (a) 9.2 cm (b) 11.9 cm (c) 8.2 cm (d) 14.8 cm (e) 7.4 cm (f) 17.7 cm. **2.** 34 m. **3.** 7.2 m. **4.** (a) 8.4 m (b) 4.7 m (c) 4.6 m (d) (b) uses the given angle and hypotenuse directly, while (c) is built on the *rounded* value of h from (a); that rounding error carries through. **5.** (a) 9.8 cm (b) 10.1 mm (c) 136.1 in (d) 5.3 ft. **6.** (a) 6.8 cm (b) 5.8 cm (c) 5.3 cm. **7.** (a) No side length is given, so there is nothing to multiply or divide by. (b) The triangle is not right-angled, so the sine, cosine and tangent ratios do not apply. (c) No acute angle is given — with two sides known, use the Pythagorean theorem instead ($x = 12.0$). **8.** (a) $ST = 35.3$ mm (b) $GI = 12.3$ cm. **9.** $x = 11.2$ m. **10.** 30.7 m. **11.** $XY = 8.1$ cm, $YZ = 11.6$ cm. **12.** D. **13.** 158. **14.** 81.

Lesson 4 Assignment

1. (a) 55° (b) 67° (c) 42° (d) 24° (e) 60° (f) 45° (g) 60° (h) 45°. **2.** (a) 36.4° (b) 78.0° (c) 39.4° (d) 77.3° (e) 87.7° (f) 51.1°. **3.** (a) 28° (b) 77° (c) 44°. **4.** (a) 24.8° (b) 58.7° (c) 35.2°. **5.** (a) 62° (b) $x = 32^\circ$, $y = 58^\circ$. **6.** (a) $AD = 35$ (b) 23°. **7.** 146°. **8.** 30.9°. **9.** (a) 43° (b) 101°. **10.** 124°. **11.** 6°. **12.** 5.39°. **13.** B. **14.** 52. **15.** 36.4. **16.** 38.1.

Lesson 5 Assignment

1. (a) 61° (b) 22.5 (c) 47° (d) 26.9 (e) 169.1 (f) 30°. **2.** 21 m. **3.** (a) 55° (b) 23.2 km. **4.** shorter side 26.4 cm, longer side 32.6 cm. **5.** 102 m. **6.** 52.6 cm. **7.** $\angle A = 49^\circ$, $AC = 17.0$, $AB = 26.0$. **8.** $\angle X = 57^\circ$, $\angle Z = 33^\circ$, $XZ = 26.6$. **9.** $\angle L = 47^\circ$, $JK = 150$ mm, $KL = 140$ mm. **10.** No side length is given — only the three angles — so the size cannot be determined; infinitely many similar triangles share these angles. **11.** A. **12.** 34.3. **13.** 38.1. **14.** 526.

Lesson 6 Assignment

1. 1058 m. **2.** (a) An angle inscribed in a semicircle is a right angle. (b) 7.0 cm. **3.** (a) 92 m (b) 27°. **4.** (a) see diagram (b) 50.5° (c) 18.5 m. **5.** 76 km/h. **6.** 20.2 m. **7.** (a) 39.1 m (b) 49 s. **8.** 13.6. **9.** (Extension) (a) angle $NBV = 48^\circ$, angle $MVK = 112^\circ$ (b) 31.4 km (c) 2.8 km (d) 31.5 km (e) 085°. **10.** (Extension) B. **11.** (Extension) D.

Lesson 7 Assignment

1. $XW = 6.0$ cm; $YW = \frac{6.0}{\tan 50^\circ} = 5.0346$; $WZ = \frac{6.0}{\tan 35^\circ} = 8.5689$; $YZ = 13.6$ cm. **2.** (a) 19.8 cm (b) 2.2 cm (c) 37.9 cm. **3.** (a) 64° (b) 42°. **4.** (a) The angles are measured from eye level, so trigonometry gives only the rise above eye level; the 1.75 m below eye level must be added back (b) 4.57 m and 5.75 m. **5.** 151.8 m. **6.** 38 m. **7.** (a) 25 m (b) runner A (22.2 s versus 24.6 s). **8.** 28 m. **9.** 68. **10.** (Extension) 9°. **11.** (Extension) B.

Lesson 8 Assignment

1. (a) opposite $E = DF = 8$, adjacent $E = DE = 6$ (b) $\sin E = \frac{4}{5}$, $\cos E = \frac{3}{5}$, $\tan E = \frac{4}{3}$ (c) $\cos F = \frac{4}{5}$. **2.** (a) 10.5 cm (b) 17.9 cm. **3.** (a) 9.1 m (b) 17.8 m. **4.** 35°. **5.** 59.8°. **6.** 54.1°. **7.** 122.0 m. **8.** (a) $AD = 24$ (b) 69°. **9.** 45°. **10.** A. **11.** 555. **12.** 94.5. **13.** $PR = 34$ m; triangle QRT supplies only $\angle TQR = 90^\circ$ and $QR = 16$ — one angle and one side cannot fix a second angle, and the missing side (QT or TR) is set by $PT = TR$, which lies outside triangle QRT; $\angle QPR = 28^\circ$, $\angle QRT = 34^\circ$; S is the midpoint of PR because $\triangle PTR$ is isosceles, $PT = 19.3$ m. **14.** (a) 0.6018 (b) 0.3256 (c) 1.3764 (d) 31° (e) 78° (f) 73°. **15.** (a) $\angle B = 52.0^\circ$, $BC = 13.3$ cm, $AB = 21.6$ cm (b) $DF = 15.8$ m, $\angle D = 53.5^\circ$, $\angle F = 36.5^\circ$. **16.** 30.5 m. **17.** (Extension) (a) 6.6 km north, 12.4 km east (b) $HN = 12.7$ km, bearing 103°. **18.** (Extension) (a) 13.0 m (b) 31.6°.